SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I P. Goutham Reddy (192525139) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

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APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Ans: **1. Scenario: Event Communication Security (Caesar Cipher)**

a. Encrypt "MEET AT NOON" using a shift of 4

Each letter is shifted **4 positions forward** in the alphabet.

Plaintext M E E T A T N O O N

Shift +4   Q I I X E X R S S R

Encrypted Message:

QIIX EX RSSR

b. How to retrieve the original message

The receiver knows that the message was encrypted with a **shift of 4**.

- Shift every letter **4 positions backward**.
- Example:
  - Q → M
  - I → E
  - X → T
  - R → N

Thus,

QIIX EX RSSR → MEET AT NOON

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.
b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

Ans. **2. Scenario: Intercepted Suspicious Message**

Ciphertext:

GSRH RH Z HVXIVG

a. Most likely original message

This is encrypted using the **Atbash Cipher** (alphabet is reversed).

Decoding gives:

THIS IS A SECRET

b. Justification

- Atbash is a **monoalphabetic substitution cipher**.
- Each letter is replaced with its opposite:
  - A ↔ Z
  - B ↔ Y
  - C ↔ X
  - D ↔ W
- Example:
  - G → T
  - S → H
  - R → I
  - H → S

The decoded sentence becomes:

THIS IS A SECRET

The repeating pattern **RH** appears twice, matching the repeated word **IS**, confirming the Atbash cipher.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.
b. Explain how the repeating keyword influences each character of the ciphertext.

Ans. **3. Scenario: Secure Messaging System (Vigenère Cipher)**

Plaintext: **HELLO**

Keyword: **KEY**

a. Encryption process

Repeat the keyword until it matches the plaintext length.

Plaintext H E L L O

Keyword K E Y K E

Convert letters to numbers (A=0):

Letter	Value
H	7
E	4
L	11
L	11
O	14
K	10
E	4
Y	24
K	10
E	4

Add modulo 26:

- H + K = 7 + 10 = 17 = R
- E + E = 4 + 4 = 8 = I
- L + Y = 11 + 24 = 35 → 9 = J

- L + K = 11 + 10 = 21 = V
- O + E = 14 + 4 = 18 = S

Ciphertext: RIJVS

b. Effect of the repeating keyword

- The keyword **KEY** repeats as **KEYKE**.
- Each keyword letter determines a different shift.
- Therefore, identical plaintext letters may encrypt differently depending on the keyword letter.
- This makes the cipher much stronger than a Caesar cipher because it uses **multiple shifting values**.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.
b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Ans **4. Scenario: Suspicious Digital Asset Investigation (Steganography)**

a. Detecting hidden data

Investigators can use:

1. **LSB (Least Significant Bit) Analysis**
   - Check pixel bits for hidden information.
2. **Statistical Analysis**
   - Compare image statistics with normal images.
   - Detect unusual pixel distributions.
3. **Steganalysis Tools**
   - Examples:
     - StegExpose
     - Stegdetect
     - zsteg
4. **Metadata Examination**

- Inspect image metadata for anomalies.

b. Two techniques to improve security

1. **Encrypt the secret message before embedding**
   - Even if extracted, the message remains unreadable.
2. **Randomize embedding locations using a secret key**
   - Hide data in unpredictable pixel locations.
   - Makes steganalysis much more difficult.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Ans **5. Scenario: Legacy Encryption (Rail Fence Cipher)**

Ciphertext:

WECRLTEERDSOEEFEAOCAIVDEN

a. Original message

This is a **3-rail Rail Fence Cipher**.

After decryption:

WE ARE DISCOVERED FLEE AT ONCE

b. Why is it vulnerable?

- It only rearranges letters; it does **not** substitute or encrypt them mathematically.
- Letter frequencies remain unchanged.
- Attackers can easily try different rail numbers.
- Modern computers can break it quickly using brute force.

c. One practical enhancement

Use **AES (Advanced Encryption Standard)** instead of only a transposition cipher.

Advantages:

- Strong encryption.
- Resistant to brute-force attacks.
- Widely used for secure communication.
- Suitable for modern corporate systems.

Summary

Question	Answer
1(a)	**QIIX EX RSSR**
1(b)	Shift each letter backward by 4 to obtain **MEET AT NOON**
2(a)	**THIS IS A SECRET**
2(b)	Identified using **Atbash substitution cipher** and repeating patterns
3(a)	**HELLO + KEY → RIJVS**
3(b)	Repeating keyword changes the shift for each character
4(a)	Use LSB analysis, steganalysis tools, statistical analysis, and metadata inspection
4(b)	Encrypt data before hiding it and use randomized embedding with a secret key
5(a)	**WE ARE DISCOVERED FLEE AT ONCE**
5(b)	Vulnerable because it only rearranges letters and is easy to brute-force
5(c)	Replace with **AES encryption** for modern security

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation